



Attention and perceptual-cognitive training:

Preliminary evidence for training attention-deficit populations

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Intro

Timeline

Groups

NeuroTracker Trial Sequence

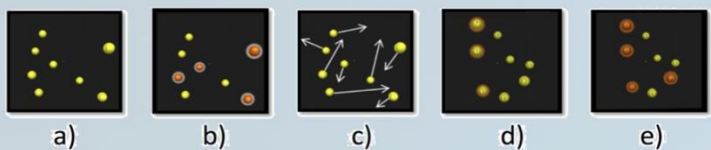
NeuroTracker is a perceptual-cognitive training system based on multiple object tracking (MOT) in a 3-dimensional (3D) virtual environment. Previous research has shown a strong link between athletic performance levels and learning capacity on this task (Faubert & Sidebottom, 2012; Faubert 2013) and transfer of this training to socially relevant tasks in aging populations (Legault & Faubert, 2012; Legault, Allard & Faubert, 2013). This preliminary study examines the training effects of 3D-MOT training on older children and adolescents with deficient attention.

Phase	Time	Description
Initial testing	Week 1	Administration of IVA+Plus
Training	Weeks 2-5	2-3 sessions per week over 4 weeks (7 sessions total)
Final Testing	Week 6	Administration of IVA+Plus

Intervention	Description
1. NT (n=17)	CORE NeuroTracker training
4. CON (n=8)	Control group. No training; equivalent wait period.

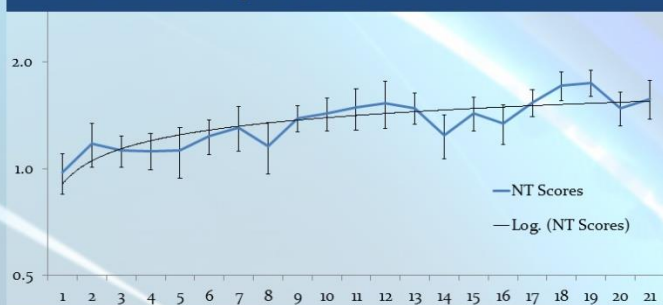
Sequence

- Presentation:** in a 3D cube eight spheres appear, coloured homogeneously in yellow. (2 seconds)
- Indexing:** the three target spheres appear in red, with a white halo. (2 seconds) Pause: the three target spheres return to their original colour and restore homogeneity. (2 seconds)
- Movement:** all eight spheres move along linear paths. If a sphere comes in contact with another sphere or a wall of the cube, it bounces off and assumes a new trajectory. (8 seconds)
- Identification:** all eight spheres cease movement and are labeled with numbers (1 to 8). Subjects verbally state their responses.
- Feedback:** target spheres are revealed, and feedback is given. (2 seconds)



If all three target spheres are correctly identified, the speed of movement increases for the subsequent trial. If not, the speed of movement decreases for the subsequent trial. Each series comprises 20 trials, three series were completed per session. At the end of each series, a threshold (average speed of correct tracking) is given.

NeuroTracker Speed Thresholds – Geometric Means

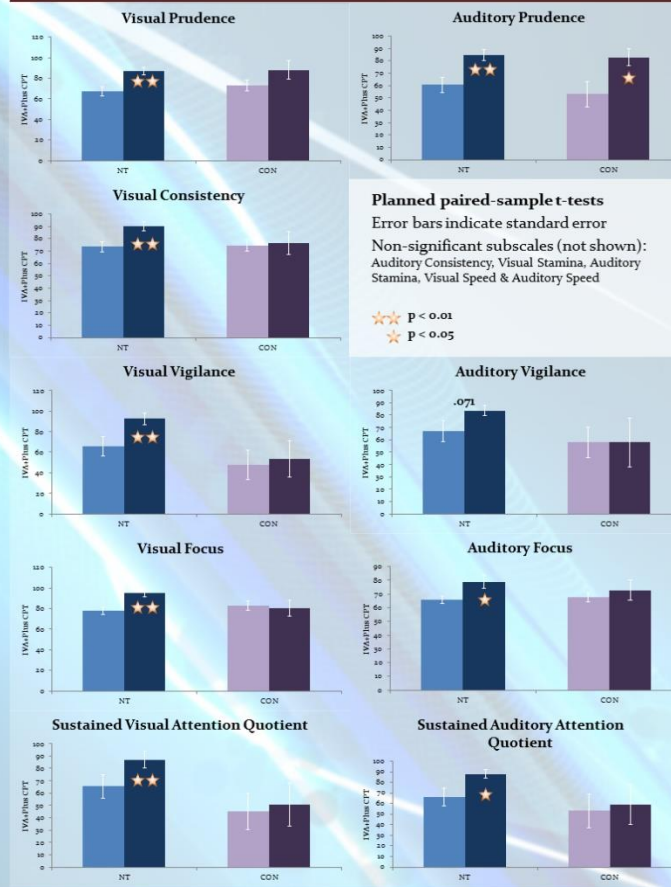


IVA+Plus Subscales		
Category	Subscale	Correlated Behavior
Response control	Prudence	Impulsiveness
	Consistency	Attentional “drifting off”
	Stamina	Sustained effort
Attention	Vigilance	Inattention
	Focus	“Tuning out”
	Speed	Information processing speed
Composite	Sustained Attention Quotient	Sustained attention

IVA+Plus Score Breakdown

Descriptive Label	Score
Exceptional	130 and above
Superior	120-129
Above Average	110-119
Average	90-109
Slightly Impaired	85-89
Mildly Impaired	80-84
Mildly to Moderately Impaired	76-79
Moderately Impaired	72-75
Moderately to Severely Impaired	68-71
Severely Impaired	61-67
Extremely Impaired	60 and below

Results



Conclusion

These findings indicate that, using a novel training paradigm (NeuroTracker perceptual-cognitive training), older children and adolescents showed **statistically significant improvements in attention and response control**. Subjects showed **improvement in both auditory and visual attention**, despite the fact that training stimulation is limited to the visual modality. Further research is needed to confirm these results and expand on the measures used.

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